PROBLEM CL10

Ventilation Device: Velocities, Force and Fan Power

Original statement · CL10.pdf

Fan | Bernoulli | Linear momentum | Numerical values

OFFICIAL PROBLEM STATEMENT

The ventilation device shown in the figure of CL10.pdf takes air out of a room at pressure p_a through a circular nozzle of diameter D_e. The air is discharged to the open atmosphere, also at pressure p_a, through a pipe of diameter $D_s < D_e$. To force the air motion the device includes a fan, which exerts a rate of work per unit time $\dot{W}$ on the air stream. To simplify the solution you may assume that the air density ρ and viscosity μ are constant and that gravity has a negligible effect.

- Using the continuity equation, obtain a relationship between the values of the velocity at the entrance and exit sections U_e and U_s, assumed to be uniform.
- Make use of Bernoulli's equation upstream from the entrance section (i.e., along a stream line between (0) and (e)) to obtain an equation for the pressure and velocity at the entrance section p_e and U_e.
- Determine the two components of the force F_x and F_y acting on the device. Include in the computation the contribution of the stagnant surrounding air.
- Use the energy equation to compute the value of U_s as a function of the power $\dot{W}$. Assume that the air temperature has the same values at the entrance and exit sections and that the device is thermally insulated.
- Obtain the numerical values of U_s, U_e, p_e, F_x and F_y for $D_e = 2$ m, $D_s = 0.1$ m, $\rho = 1.2$ kg/m^3, $\mu = 1.8 \times 10^{-5}$ kg/(m·s), $\dot{W} = 1000$ W, $p_a = 101300$ Pa. Note: the following quantities are known: $D_e, D_s, \rho, \mu, \dot{W}, p_a$.

Source: CL10.pdf (page 1). Notes.pdf: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.35, integral total energy balance Eq. 3.46, Bernoulli's equation Eq. 4.17.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady, constant ρ and μ; neglect $\bar{g}$; uniform properties at e and s; same temperature at e and s ($e_e = e_s$); device insulated.
- The air in the room far from the inlet is at rest at p_a (point 0); the exit is a free jet at p_a; the outer surface of the device is surrounded by stagnant air at p_a (included through the gauge pressure).
- Geometry (figure): the inlet faces downwards, air enters along $+\bar{e}_y$; the exit pipe points along $+\bar{e}_x$.
- Data: $[D_e,\ D_s,\ \rho,\ \mu,\ \dot{W},\ p_a]$. Unknowns: $[U_e,\ U_s,\ p_e,\ F_x,\ F_y]$.

Symbol	Meaning	SI unit
D_e, D_s	inlet and exit diameters	m
U_e, U_s	inlet and exit velocities	m/s
p_e, p_a	inlet and ambient pressures	Pa
$\dot{W}$	fan power given to the air	W
F_x, F_y	force on the device	N

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the device: $\bar{e}_x$ horizontal (exit direction), $\bar{e}_y$ vertical upwards (inlet flow direction).
- Surfaces: Σ_e (inlet, $\bar{n} = -\bar{e}_y$, $\bar{v} = U_e \bar{e}_y$), Σ_s (exit, $\bar{n} = +\bar{e}_x$, $\bar{v} = U_s \bar{e}_x$), Σ_w (fixed walls) and Σ_m (fan).

$$\boxed{\bar{F} = F_x \bar{e}_x + F_y \bar{e}_y = \int_{\Sigma_w, \Sigma_m} \left[(p - p_a)\, \bar{n} - \bar{\bar{\tau}}' \cdot \bar{n} \right] d\sigma}$$

PHASE 3

Balances, Surface by Surface

(1) Mass (Notes Eq. 3.11)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c}\rho\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\,(\bar{v}-\bar{v}_c)\cdot\bar{n}\,d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,(\bar{v}-\bar{v}_c)\cdot\bar{n}\,d\sigma$
Σ_e	$-\rho U_e \frac{\pi D_e^2}{4}$
Σ_s	$+\rho U_s \frac{\pi D_s^2}{4}$
$\Sigma_w,\ \Sigma_m$	0

$$\boxed{U_e\,D_e^2 = U_s\,D_s^2} \qquad (1)$$

(2) Bernoulli from the room (0) to the inlet (e)

$$p_a + 0 = p_e + \tfrac{1}{2}\rho U_e^2 \implies \boxed{p_e = p_a - \tfrac{1}{2}\rho U_e^2} \qquad (2)$$

(3) Linear momentum (Notes Eq. 3.35)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c}\rho\bar{v}\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c}(p-p_a)\,\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma + \underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma$	$-\int (p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_e	$\rho U_e\bar{e}_y(-U_e)\frac{\pi D_e^2}{4} = -\rho U_e^2\frac{\pi D_e^2}{4}\bar{e}_y$	$-(p_e-p_a)(-\bar{e}_y)\frac{\pi D_e^2}{4} = (p_e-p_a)\frac{\pi D_e^2}{4}\bar{e}_y$	0
Σ_s	$+\rho U_s^2\frac{\pi D_s^2}{4}\bar{e}_x$	0 (free jet)	0
$\Sigma_w,\ \Sigma_m$	0	$-\int (p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed terms: $-\bar{F}$.

$$\boxed{F_x = -\rho U_s^2\frac{\pi D_s^2}{4}, \qquad F_y = \rho U_e^2\frac{\pi D_e^2}{4} + (p_e-p_a)\frac{\pi D_e^2}{4} = \rho U_e^2\frac{\pi D_e^2}{8}} \quad (3)$$

The last form of F_y uses (2): $(p_e - p_a) = -\frac{1}{2}\rho U_e^2$.

(4) Total energy (Notes Eq. 3.46)

$$\underbrace{\frac{d}{dt}\int_{V_c}\rho\left(e+\tfrac{v^2}{2}\right)dV}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\left(e+\frac{v^2}{2}\right)[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c} p\,\bar{v}\cdot\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma + \underbrace{\int_{V_c}\rho\bar{g}\cdot\bar{v}\,dV}_{\text{neglect }\bar{g}} - \int_{\Sigma_c}\bar{q}\cdot\bar{n}\,d\sigma + \underbrace{\int_{V_c}Q_R\,dV}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\left(e+\frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$	$-\int \bar{q}\cdot\bar{n}\,d\sigma$
Σ_e	$-\dot{m}\left(e_e+\frac{U_e^2}{2}\right)$	$+p_eU_e\frac{\pi D_e^2}{4}$	0	0
Σ_s	$+\dot{m}\left(e_s+\frac{U_s^2}{2}\right)$	$-p_aU_s\frac{\pi D_s^2}{4}$	0	0
Σ_m	0	$-\int_{\Sigma_m} p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int_{\Sigma_m}\bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$	0
Σ_w	0	0	0	0 (insulated)

Boxed Σ_m terms: the fan power $\dot{W}$. With $e_e = e_s$, $\dot{m} = \rho U_s\pi D_s^2/4$ and (2), $\frac{p_a-p_e}{\rho} = \frac{U_e^2}{2}$:

$$\dot{W} = \dot{m}\left[\frac{p_a-p_e}{\rho} + \frac{U_s^2-U_e^2}{2}\right] = \dot{m}\,\frac{U_s^2}{2} \implies \boxed{\dot{W} = \rho\,\frac{\pi D_s^2}{8}\,U_s^3, \qquad U_s = \left(\frac{8\dot{W}}{\rho\pi D_s^2}\right)^{1/3}} \quad (4)$$

(5) Numerical values

Quantity	Expression	Value
U_s	$\left(8\dot{W}/(\rho\pi D_s^2)\right)^{1/3}$	59.647 m/s
U_e	$U_s\,(D_s/D_e)^2$	0.14912 m/s
p_e	$p_a - \frac{1}{2}\rho U_e^2$	101299.987 Pa ($p_a - p_e = 0.0133$ Pa)
F_x	$-\rho U_s^2 \pi D_s^2/4$	-33.531 N
F_y	$\rho U_e^2 \pi D_e^2/8$	0.04191 N

PHASE 4

Interpretation, Limits and Dimensional Check

- The large intake makes the inlet velocity tiny ($U_e \approx 0.15$ m/s), so the suction depression $p_a - p_e$ and F_y are negligible; the force is the reaction to the exit jet, $F_x < 0$ (opposite to the jet).
- The Reynolds numbers, $\rho U_s D_s/\mu \approx 4\times10^5$ and $\rho U_e D_e/\mu \approx 2\times10^4$, are large, which supports neglecting viscous stresses on the sections and using Bernoulli upstream of the inlet (this is where μ enters).
- Dimensions (SI): $[\rho D_s^2 U_s^3] = (\mathrm{kg/m^3})\,\mathrm{m^2(m/s)^3} = \mathrm{W}$; $[\rho U^2 D^2] = \mathrm{N}$.
- Agreement with the official solution: CL10.pdf obtains (1) to (4), with $F_x = -\rho U_s^2\pi D_s^2/4$, $F_y = \rho U_e^2\pi D_e^2/4 + (p_e - p_a)\pi D_e^2/4$ and $\dot{W} = \rho\pi D_s^2 U_s^3/8$; it does not write out the numbers of part 5.